

Quantum Reference Guide

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This reference is a topic-based index that links to the in-depth tutorial sections. Use it to jump directly to a subject and then explore details in context.

Chapter 1. Platform

- Overview: [Platform overview](#)
- Project setup: [Project setup](#)

Chapter 2. Modeling and Tenancy

- Ontologies in Quantum (deep dive): [Modeling relationships with ontologies](#)
- Concrete e-commerce example: [Orders, Shipments, Fulfillment, Returns](#)
- Integration with Morphia/permissions: [Integrating ontology with rules and repos](#)
- Domain modeling: [Domain modeling fundamentals](#)
- Dynamic edge properties and facet segmentation: [Open edge properties, edge filtering, and facet segmentation](#)
- Completion tasks and groups: [CompletionTasks, CompletionTaskGroups, payloads, and event stream](#)
- Tenant models and realms: [Multi-tenant model and realm separation](#)
- User-definable facet segmentation & coordinate SPI: [Multi-tenant coordinate SPI and facet resolution](#)
- Tenant onboarding and access invites: [Provision tenants, configure onboarding workflow, and invite users into a tenant](#)
- Domain rule context: [Domain rule context](#)

Chapter 3. APIs and Queries

- REST CRUD: [Building RESTful CRUD APIs](#)
- Query language: [Query language and filtering](#)
- Planner and QueryGateway: [Governed aggregation pipelines, multi-hop traversal, and REST endpoints](#)
- Relationship hydration: [Relationship hydration with expand\(path\)](#)
- Scoped action enablement: [Why an action is unavailable: policy, feature flags, settings, and readiness blockers](#)
- AI Agent integration: [AI Agent Integration](#) (tools discovery, schema, execute, session/trace)
- MCP Server and Client: [MCP Server and Client](#) (Claude/Cursor integration, ontology tools, end-to-end example)

Chapter 4. Security

- Authentication: [Authentication, permissions, and annotations](#)
- JWT tokens for API access: [Generating JWT Tokens](#) (login, service tokens, key pair generation)
- Service tokens for integrations: [Service Tokens](#) (MCP servers, cron jobs, CI/CD, partner APIs)
- Multiple auth providers: [Multiple Authentication Providers](#) (issuer resolution, per-tenant key pairs, external identity providers)
- Key pair management: [Key Pair Management](#) (default keys module, configurable paths, production setup)
- Shared links: [Shared Links](#) (time-bound link sharing, HMAC signing, access logging, revocation)
- Comment chains and media: [Comment Chains and Media References](#) (hierarchical discussion, governed object metadata, short-lived grants)
- OAuth server: [OAuth Server Module](#) (built-in OAuth 2.0/OIDC authorization server, ChatGPT/Gemini/Grok integration, no external IdP needed)
- Permissions: [Permissions](#)
- Security annotations: [Security annotations](#)
- Permission Resource Check APIs: [Permission Resource: Check APIs and Client Usage](#)
- Label resolution and script helpers: [Labels \(SPI\) and hasLabel\(\)](#)
- Secrets and vault (global vs tenant-level): [Secrets and Vault Configuration](#)

Chapter 5. Data Seeding

- Seed packs (tutorial): [Seed packs: Declarative tenant seeding](#)
- Seed packs (deep dive document): [Seed packs and declarative tenant seeding](#)
- Bootstrap packs: [Bootstrap Packs](#) (runtime orchestration, step handlers, startup application, admin APIs, and when to choose them over seed packs)

Chapter 6. Email and Notifications

- Email templates: [Email Templates](#) (Qute rendering, classpath defaults, realm overrides, preview APIs, onboarding template keys, Postmark delivery)

Chapter 7. Operations

- Migrations: [Migrations](#)
- Testing: [Testing](#)
- Usage metering and token allocation: [Usage Metering and Token Allocation](#) (API/LLM usage, token pools per tenant, scoped to APIs and Tools/LLM configs)

Chapter 8. Samples

- Supply Chain tutorial: [Supply Chain](#)

Chapter 9. Appendix

For configuration, Quarkus integration details, and glossary, see the Appendix in the main index.